IIT Guwahati
North Guwahati 781039, Assam, India

Technothlon

the international school championship

.....Inspiring Young minds!

PAPER THEME

HAUTS SQUAD

Techniche
The annual Techno-Management Festival
IIT Guwahati

General Instructions

- This question paper consists of 5 sections with a total of 18 questions (as this is the 18th edition of Technothlon).
- There is a bonus section, in the end, you will get to know about it as you go through the question paper.
- The marking scheme is clearly given below and it is one of the most unique schemes you might ever experience.
- There are two sections (Section 4) where you have to select the marking scheme according to your choice in the first question (Q.0) of the section. Failing to opt the marking scheme of your choice will result in selection of the marking scheme in the first option by default.
- No ambiguity related to the marking scheme will be entertained later.
- If any discrepancy in the paper is found, the final decision regarding the same will be taken by Team Technothlon itself.
- Integer type answers can have any integer value (NOT restricted to 0 to 9)

Marking Scheme

- **Choose your ally:**

The most interesting marking scheme you might have seen. Out of the given two options; choose the marking scheme that you think is best for you. (You can check the questions before marking your choice)

- **Power sequence:**

Power Scheme: With negative marking
For consecutive correct questions, you will be awarded 20, 21, 22, 23 and so on.
However, if you leave a question or answer it incorrectly, the sequence is broken, and you will start
again from 1.
If you answer the question incorrectly your marks will be deducted in the same sequence
E.g. Solving 4 questions,

For all correct you get 1+2+4+8 = 15
For all incorrect you get -1-2-4-8 = -15
For RRWW you get 1+2-1-2= 0
For RRRW you get 1+2+4-1 = 6
For RRWR you get 1+2-1+1 = 3
For RRUR you get 1+2+0+1=4

And so on.
(R -> right answer, W-> wrong answer, U-> Unattempted)

- **Affinity in a row:**

Q1, Q2, Q3 have marks 2, 4 and 6 respectively **BUT**
There are two conditions:

1. There would be no reduction in the marks of this section if you solve 2 consecutive questions correctly, also
2. If the above criteria is not met alongside you solving a question incorrectly the total marks you have scored in this section are halved.

For example :
For a 3 question section-
If you solve Q2 and Q3 correctly but Q1 incorrectly or leave it unattempted then you will get 0 + 4 + 6 = 10 marks.
If you solve Q1 and Q3 correctly but Q2 incorrectly or you have not attempted it then you will get (2 + 0 + 6)/2 = 4 marks.
Similar rules are applied for a section containing 4 questions with the last question having 8 marks.

- **Penaltyless Victory**

Answering a Question correctly Gives You 5 marks but no marks are deducted for incorrect answers. (+5,0)

- **Good ol' Scheme**

Answering a question correctly gives you 5 marks, whereas answering incorrectly reduces 2 from the total marks you gain. (+5,-2)

- **Shield me**

You get 5 marks for answering correctly and 4 marks are deducted for answering incorrectly, but if you have solved the previous question correctly then there is no negative marking for the current question.
For example:
If Your sequence is RWRW then your score is 5+0+5+0 = 10
But if your sequence is RWWR then your score would be 5+0-4+ 5 = 6
And if your sequence is RWUW then your score is 5+0+0-4 = 1
Where (R -> right answer, W-> wrong answer, U-> Unattempted)

- **Risk-Reward**

You get 6 marks for answering any question correctly and none are deducted for an incorrect answer, but if you have answered any question incorrectly your marks for this whole section is halved.
For example: In a section consisting of 4 questions if you solved any two correctly and any one of the other two or both incorrectly you will get 6 marks.

SECTION 1:

Marking Scheme: Affinity in a row

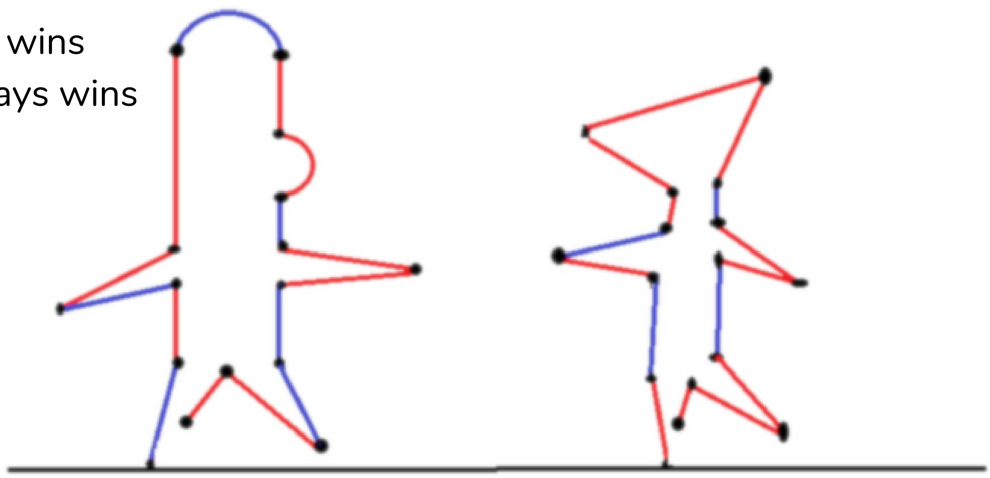
Maximum marks: 12
Minimum marks: 0

Phineas and Ferb are two primary school kids living in the City of Danville. But don't confuse them with the other kids in the neighborhood, they're unique. They seek ways to occupy their time during their summer vacation. Often these adventures involve elaborate, life-sized, and seemingly dangerous engineering projects. Phineas' older sister **Candace** has two obsessions: exposing Phineas and Ferb's schemes and ideas, and winning the attention of a boy named Jeremy.
Meanwhile, the boys' pet **platypus Perry**, acts as a secret agent for an all-animal government organization called the O.W.C.A. ("Organization Without a Cool Acronym"), fighting the evil **Dr. Heinz Doofenshmirtz.**

1. [MULTIPLE CORRECT ANSWER]

Phineas and Ferb are sitting under a tree, bored, drawing and wondering what to do. Meanwhile, Ferb admires their drawing as they drew their own figures in 2 colors red and blue, giving Phineas an idea for a game.
The game is as follows -
On his turn, a player "cuts" (erases) any line segment of his choice but of one particular color only. All the line segments are no longer connected to the ground by any path "falls" (i.e., gets erased). The first player who is unable to move loses.
Who do you think will win the game, if both players play optimally and Ferb chooses the color Blue and Phineas Chooses the color Red?

a) The First Mover always wins
b) The Second Mover always wins
c) Blue Always wins
d) Red always Wins



Ferb was rejected from the school's basketball team as his teacher dismissed him for being too weak to play. So he, along with Phineas, thinks about conducting a tournament of his own to show the world that the so-called "weak" can also win if they're given a chance.

2. [INTEGER TYPE ANSWER]

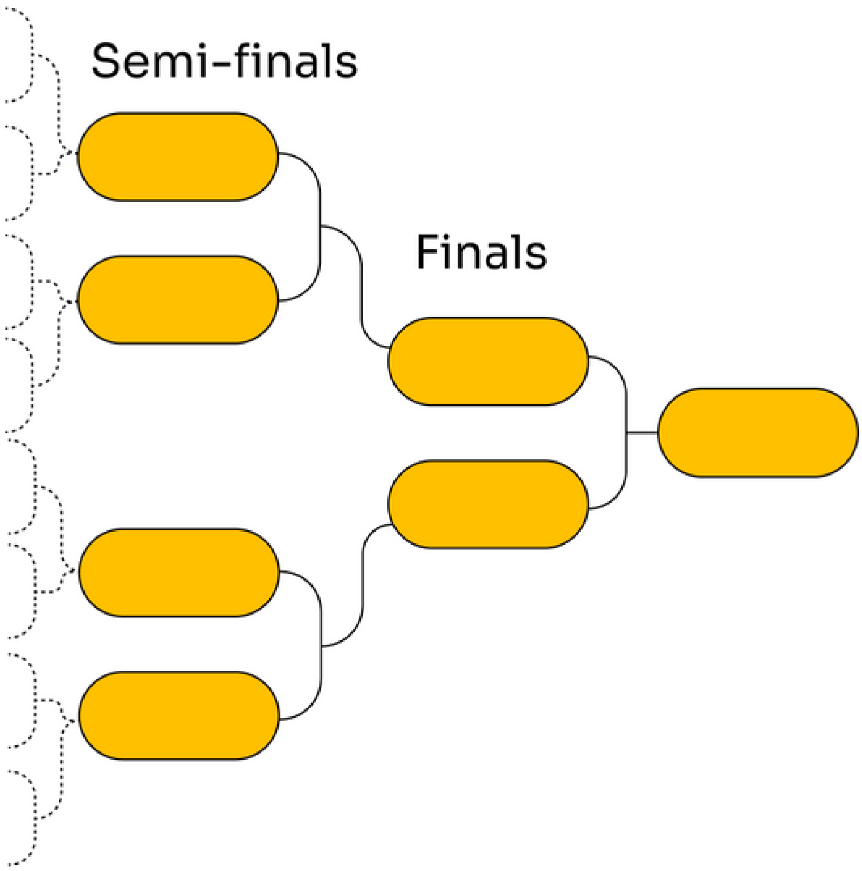
Phineas and Ferb developed 64 distinct AI robots and ranked them in decreasing order of strength as 1,2,3... Up to 64 (Rank 1 is the strongest and Rank 64 is the weakest) and puts them against each other in an elimination format.

A robot who is ranked 3 or more places higher (according to strength) than its opponent will definitely win, while either of the robots can win when the two robots playing against each other have a "close encounter"(they have a rank difference of 2 or less).

It's a simple elimination type with Round of 64 initially; then Round of 32; Round of 16 (pre-quarters); Round of 8 (quarters); semis and the finals.

Now your work is to create the match-ups (draws) such that the weakest robot (robot having the highest rank in numbers) can win the tournament.

So, find out the rank of the weakest robot that can win the tournament assuming that luck is always in your favor and the weakest robot that you choose wins all his close encounters.



After successfully conducting this tournament that contains such a beautiful message for youngsters out there, their spirits are high. Now as the weekend approaches, they have a wonderful idea to bring joy to the city of Danville: "Build a Rollercoaster".....

Their friends Isabella, Baljeet, Buford, Ginger, Suzy, and Katie stop by to see what is going on. Phineas shows them their plan

3. [INTEGER TYPE ANSWER]

Phineas, Ferb, Isabella, Baljeet, Buford, Ginger, Suzy, and Katie sit at a round table to discuss the budget. Each one of them has some money(in dollars) in their hand.
They start to distribute the money among themselves in a weird manner in order to have some fun. Firstly Phineas gave all others as much money as the receiver already had (Ex: Let's say Ferb had X dollars initially, so Phineas gave him another X dollars, and if Baljeet has Y dollars initially, then Phineas gives him another Y dollars).
Then Ferb decided to follow the same process. Then Isabella, and so on to the last person i.e Katie. It was found that each of the eight persons had the same amount in the end.

Find the smallest total amount that all 8 of them (cumulatively) could have had originally in order to get the above end result.

SECTION 2:

Marking Scheme: Penaltyless Victory

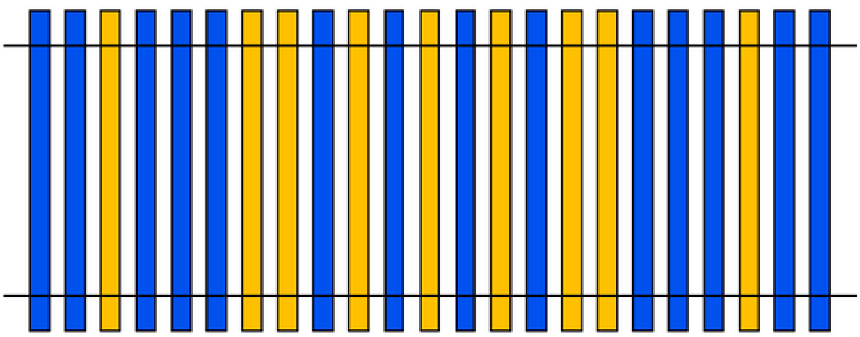
Maximum marks: 16
Minimum marks: 0

Candace is in charge of taking care of Phineas and Ferb. But to our surprise, she dozed off while keeping an eye on them. And Phineas and Ferb seizing the opportunity start their adventure of building the roller coaster

1. [MULTIPLE CORRECT ANSWER]

Phineas and Ferb have designed the launch track of yellow and blue rails but they plan to change all yellow rails to blues. But they follow certain rules to change the rails:-

Rule 1: They can change any yellow rail to a blue rail by paying 1$ or reverse the whole sequence without paying anything.
Rule 2: Reverse operation is only allowed when the current helix (stack of blue and yellow rails) is not a palindrome and also the last move was not a reverse operation
The person who pays less amount after converting all the rails to blue rails wins. If both play optimally who wins if Phineas plays the first move.
NOTE: A palindrome is a sequence of characters that reads the same backwards as forwards, such as RACECAR or 12321 or NITIN or NAMAN...



a) Phineas
b) Ferb
c) Game ends in a draw
d) Game could go on for infinite amount of time

2. [SINGLE CORRECT ANSWER]
In the making of this blue launch track, what will be the minimum amount required to pay, if they change all yellows to blues according to the rules in the above question?

a) 9
b) 4
c) 5
d) 8

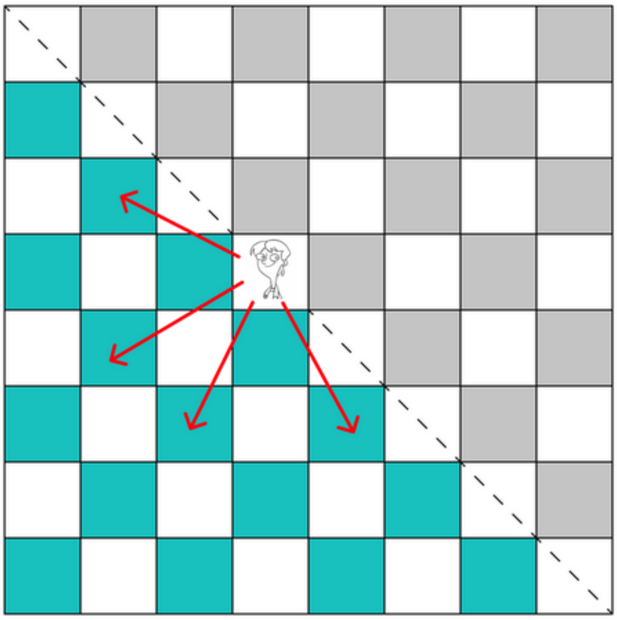
Candace is awakened by the ring of her phone. It's Stacy (Candace's best friend), wanting to go to the mall with her. But she says she can't go because she has to watch her brothers.
While Candace was talking on the phone, the boys had been transporting various building materials into the backyard. This breaks her concentration and she yells at the boys, "Will you hold it down? I am trying to use the phone!". Going back to talk to Stacy, she gets her first hint that something is wrong. Stacy tells Candace that she can see something being built in the backyard from her house. She calls over Stacy(who also brings Jeremy with her) to her house in order to investigate what the boys are doing.

3. [SINGLE CORRECT ANSWER]
Candace and Stacy both try to investigate Phineas and Ferb's doings.
They have a square-shaped backyard of size 32 meters X 32 meters. They are initially at square a1 as is shown in the figure. But only one person can move on that backyard otherwise the boys might get aware of Candace's inspection hence Stacy asked Jeremy to inspect for them.

Candace and Stacy give alternate instructions to Jeremy. He (Jeremy) can take steps only in L-shape(like a knight/horse in chess) on every instruction. After every move is played Jeremy can't move in the greyed-out squares (Squares above and containing the diagonal) i.e. On each turn, both Candace and Stacy can move Jeremy in the 4 directions shown in the figure below

Candace would start first and is allowed to start from anywhere in the backyard (except the winning square i.e. position of Phineas and Ferb obviously).
Both Candace and Stacy give optimal instructions. How many squares are winning squares for Candace, i.e Jeremy reaches Phineas and Ferb finally on her move?

a) 257
b) 767
c) 768
d) 256



As Jeremy confirms that something is fishy, Candace runs out to the backyard, then stops in her tracks, horrified at what she saw. She looks up and sees a patchwork of pipes and a launch track.

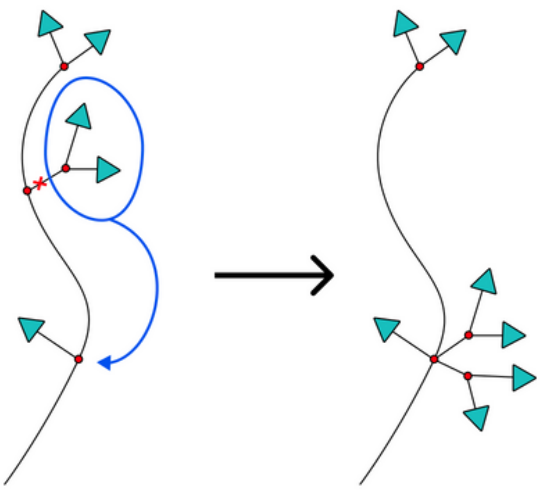
4. [SINGLE CORRECT ANSWER]

Candace shockingly asks Phineas what it is, and Phineas asks if she likes it. Whatever it is, she doesn't like it. She angrily storms off to break these patchwork of wires equipped with transmitters.

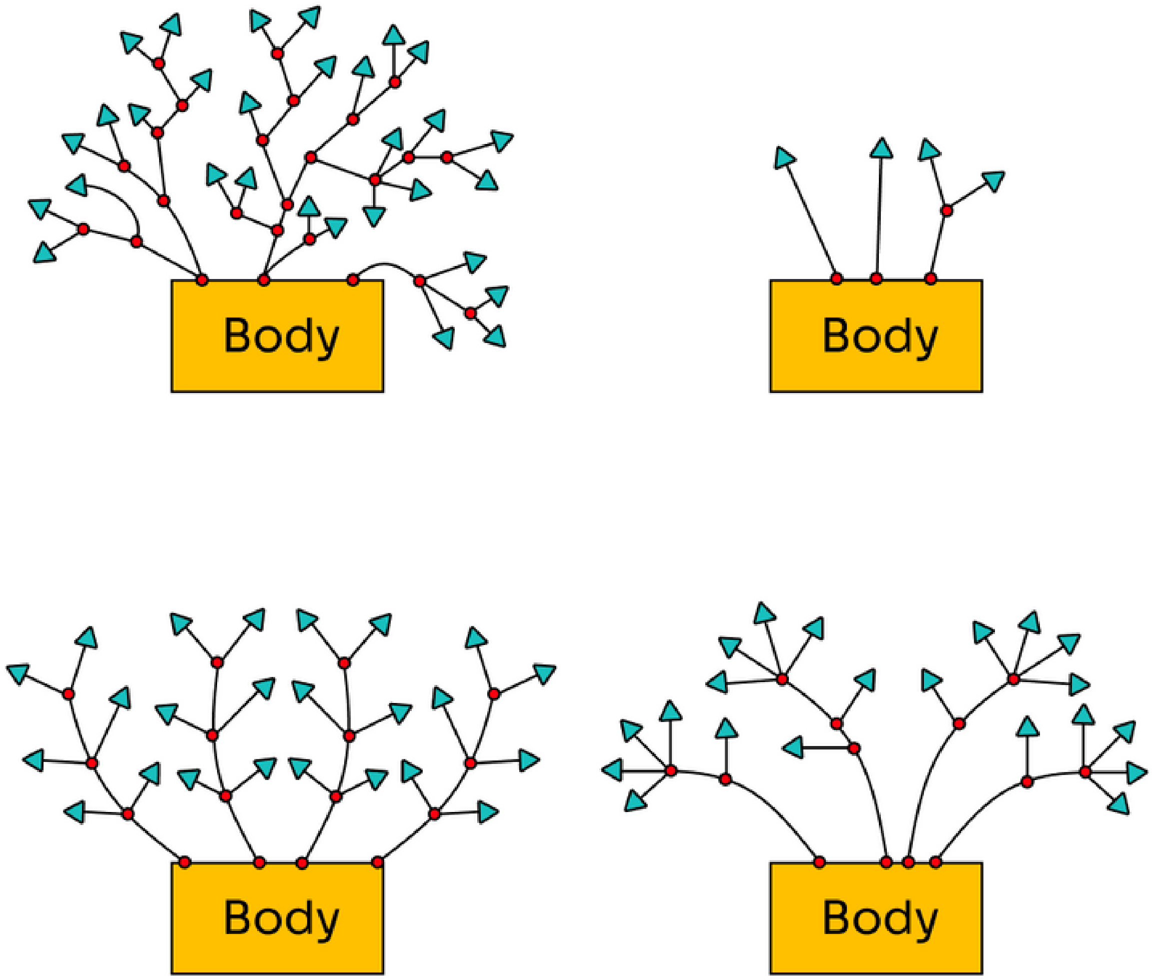
However, the wires have been sprayed with a mystical spray such that

- If she cuts off a wire attached directly to the body, the transmitter on the wire falls and nothing further happens.
- If she cuts off a wire attached to a node, new transmitters are grown as follows: look at the node from which the wire grew. Move one node closer to the body. Finally, from the node closer to the body, grow back two copies of the decapitated part that starts at the node from which the wire grew. (Example is given below)
- If a node has no "outgoing" wire, then it is equipped with a transmitter.
- The patchwork of wires falls when it has no more transmitters.

Example:



Among the given network of wires, how many can she break?



a) 0
b) 1
c) 3
d) 4

SECTION 3:

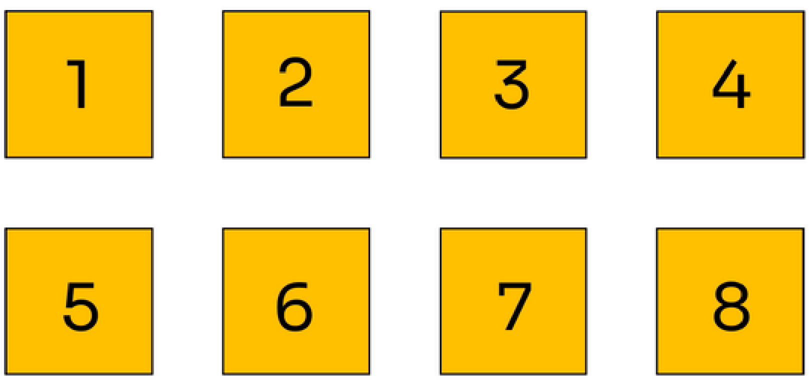
Marking Scheme: Good Ol' Scheme

Maximum marks: 15
Minimum marks: -6

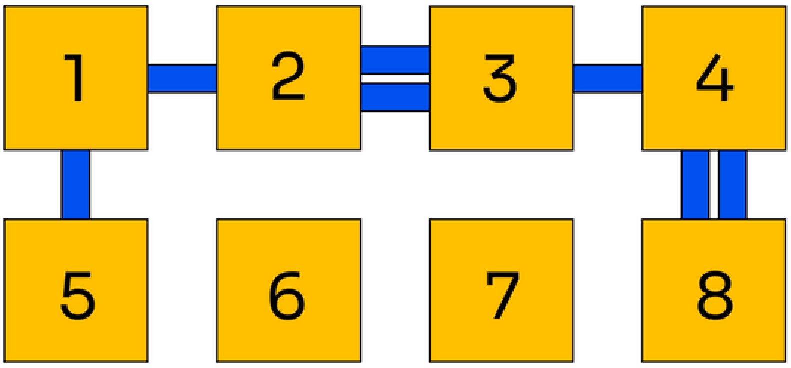
Candace realizes that this is her opportunity to expose the boys' work in front of her mother. She rides off on her bike to the supermarket.
Meanwhile, Phineas notices that their pet Perry is missing, but without giving much heed to it, goes back to complete his chores.
Perry is walking nearby alongside the house. He stops and checks to make sure no one can see him. Jumping up onto his hind legs, Perry's behavior changes. He is no longer a platypus that "doesn't do much". He puts on a brown and black hat, becoming "Agent P".

1. [**INTEGER TYPE ANSWER]**
Perry, the platypus has eight underground rooms, numbered 1 through 8; which are to be connected using bridges. But certain rules are to be followed.

- Only adjacent rooms can be connected by tunnels and there can be more than one tunnel between 'adjacent' rooms.
- 2 rooms have no tunnels connecting them with any other rooms i.e. 2 rooms are isolated These two must be 'adjacent'.
- 1 of the rooms has 1 tunnel leading from it; 2 of them have 2 tunnels leading from each and the remaining 3 have 3 tunnels leading from each.
- The six rooms which have tunnels must be connected such that each of the rooms can be reached from the other five rooms via the tunnels.



Note: 'Adjacent' are the ones which are to the left-right of each other or are above-below each other. Diagonal rooms are not considered 'adjacent'.
One possible map can be:

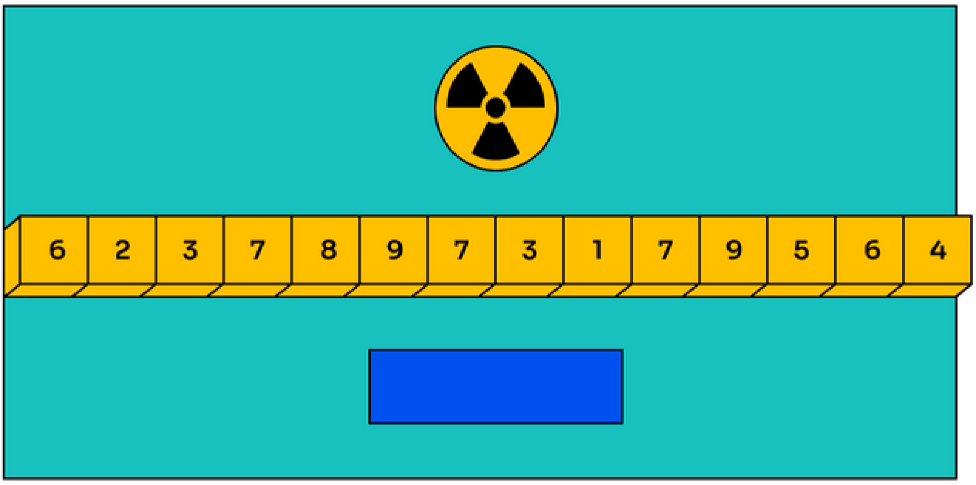


How many such map configurations exist?

A remote control opens a hidden door in the side of the house and Agent P steps in and rides the tube elevator down to his room.
Major Monogram (Agent Perry's head at the secret service organization) informs him that Dr. Doofenshmirtz has bought "80% of the country's tin foil". Agent P must find out why and stop him. Before Agent P leaves, Major Monogram reminds him to maintain his cover as a "mindless domestic pet".

2. [INTEGER TYPE ANSWER]
Hence, Agent P decides to fill up this secret room he's (Dr. Doofenshmirtz) residing in.
The operator needs to select the numbers(each once) from the keys on the machine. The machine creates the same number of cubic bricks as the number formed by serially pressing the keys. (E.g. if we press 6,2,7,9,7 The machine will produce 62797 small 1X1X1 cubes). But Dr. Doofenshmirtz knowing this tries to trick Agent P by replacing every key on the machine with a different key such that the number on the new keys is 10-n (Ten minus "n") where n was the previous number on each of the respective key. Eg: If previously a key had 7 written on it, that same key now has the number 3.

The keys must be pressed only once and serially from left to right.
A number of special boxes are kept in a row to form a huge cubic meter space.
Perry measured one of the special boxes and figured that it was 7 meters in length and 11 metres in width and had a height of 13 metres.
Perry doesn't know the exact count of boxes kept. Which combination will he require to fill the space? The cubes are highly radioactive hence if produced more than required, the entire world could be put at risk and if produced less Agent P won't be able to fill up his secret room.



Giving the Major a quick salute, he flies off in his platypus-shaped hover-jet. He exits to the surface through a tunnel. Shortly thereafter, he flies by the boys and pulls his hat down over his face to avoid being recognized.
The boys are taking a break from construction to discuss the next part of the coaster, the seating chairs. They want all the rows to be in a certain way for the internal circuit to work properly. Isabella calls this arrangement of rows "Genius-Phineas circuits".

3. [INTEGER TYPE ANSWER]
Find the number of "Genius-Phineas circuits" if the first row is given below consisting of only Yellows and Blues.

First Row Arrangement = **"Y Y B Y B B Y Y Y Y B Y Y"**

Two rows P and Q are called Genius-Phineas circuits if we can get row P from row Q by doing the following operations zero or more number of times.

Operation: Choose any sub row (a smaller string of B's and Y's) of Q which contains an even number of Blues and reverse it.

For example, for Y Y B Y B, the row Y B Y B Y is a "Genius-Phineas circuit" arrangement because it can be obtained by reversing the sub row from position 2 to 5 of the original row (i.e. Y Y B Y B).

SECTION 4:

Marking Scheme: Choose Your Ally

Option 1 - **Risk Reward**
Option 2 - **Shield me**

Maximum marks: 24
Minimum marks: -16

Q0) [SINGLE CORRECT ANSWER]
So after going through the questions in this section, which marking scheme do you want to choose?
a) Shield me   b) Risk-Reward

Phineas heads off to get the required materials while Ferb resumes welding. Meanwhile, Candace tries to convince her mom and tells her about the roller coaster. Linda thinks Phineas is a little young to be a roller coaster engineer. Nearby, Agent P swings to the ***Doofenshmirtz Evil Incorporated (DooEIn)*** *building.*
As Perry reaches DooEIn he finds himself in a conundrum, the window of the building he planned to enter with is actually closed.

1. [INTEGER TYPE ANSWER]
On the 51X51 grid wall of the building, there is (danger symbol) on one of the squares and (Heart symbol) on the other 2600 squares. In one move Agent P (Perry) may reverse the symbols (danger symbol to heart symbol and heart symbol to danger symbol) of any **a**X**a** subgrid where a>1. He has to reach a stage when each square on the grid is (Heart symbol) to open the window.
How many such places are possible for (danger symbol)?

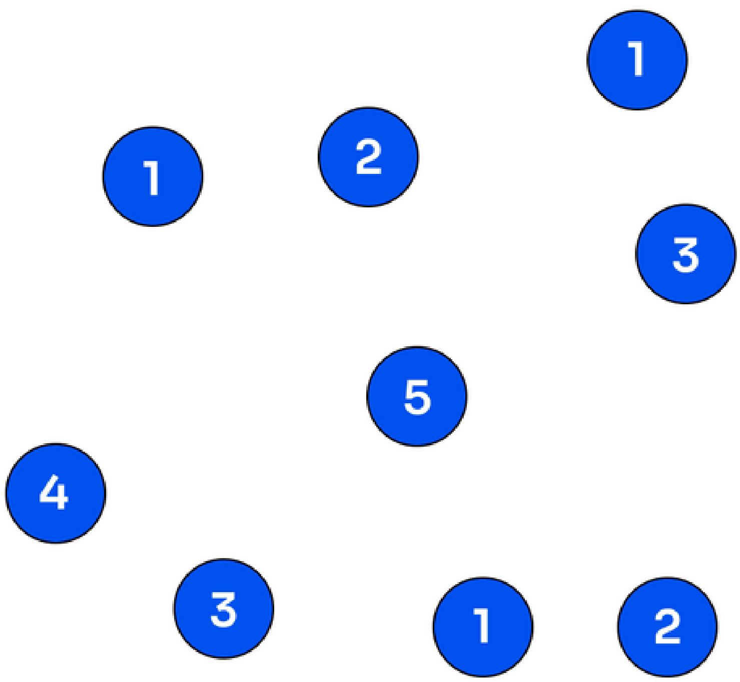
2. [INTEGER TYPE ANSWER]
How many Minimum numbers of moves will Agent P take to open the window in the above question.

Agent P swings inside Doofenshmirtz Evil Incorp. ***(DooEIn)*** *Building, but before he can do anything, Doofenshmirtz catches him using his robotic arm. "This is the new machine I have made just to stop you… you naughty Perry the Platypus"…….. "Catch-inatorrrrrr" he roars as if the whole city was listening.*
He locks up Agent P inside a cage.

Meanwhile, back at home Phineas heads off to get solid-fuel booster rockets for his rollercoaster. Coming back, he discusses the roller coaster project with Ferb.

3. [INTEGER TYPE ANSWER]
The project is to build 20 tracks connecting 9 locations in the city. So far, only some of the tracks are constructed. The project is taking a lot of time and the friends are getting impatient. Under pressure to complete the job, Phineas and Ferb count the number of tracks that are remaining to be built.

To make their job easier Phineas has left you with the above graph in which the colored circles(nodes) represent the locations and the digit on each of the locations indicates the number of constructed tracks to other locations in the city.
Find the no. of tracks that are yet to be constructed and highlight the roads which have been completed on the following graph.



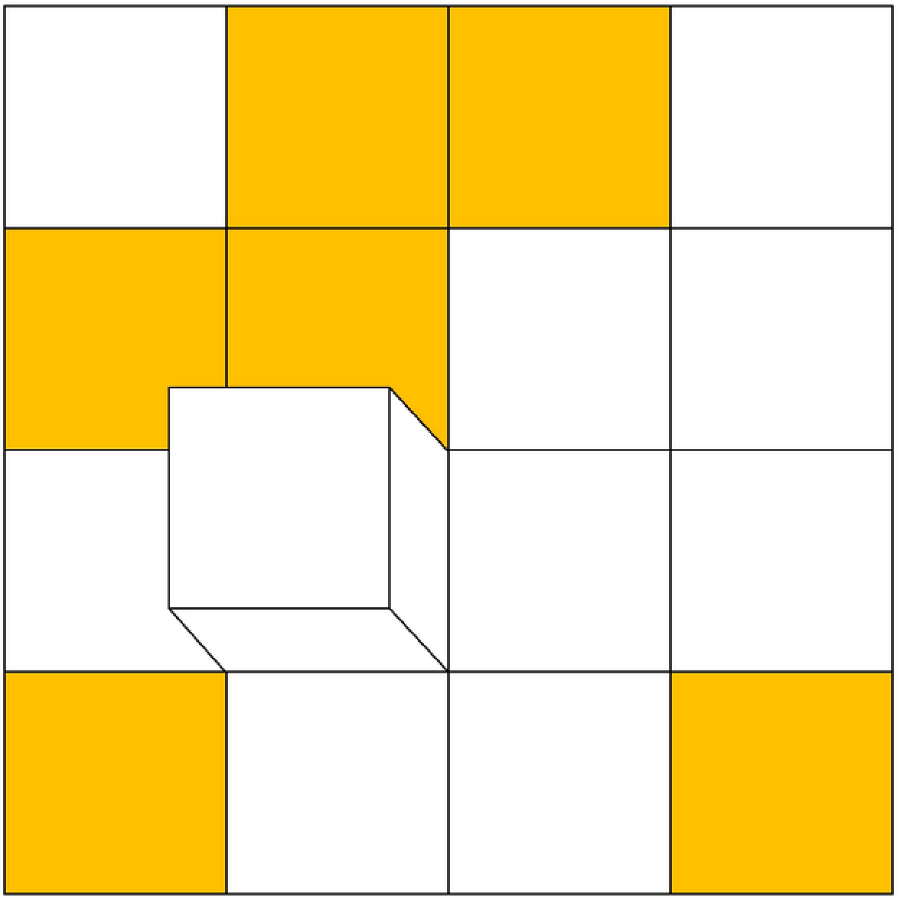
After completing the above work, you require a supporting staff's cubic block in the mid-air. Your task is to paint the block's total surface area but it can be done only by the below-given process.

4. [INTEGER TYPE ANSWER]

Roll the cubic block around the given surface, picking up the yellow squares on its faces. Try to get all the yellow squares onto the block at the same time, in as few moves as possible.

After every roll, the surface square and cube face that you brought into contact swap their colors, so that a non-yellow cube face can pick up a yellow square, but a yellow face rolled onto a non-yellow square puts it down again.

You can only roll horizontally or vertically and every 90-degree roll on the edge (which is on the ground) is considered as a move. You cannot go outside the 4X4 grid.



SECTION 5:

Marking Scheme: Power Sequence

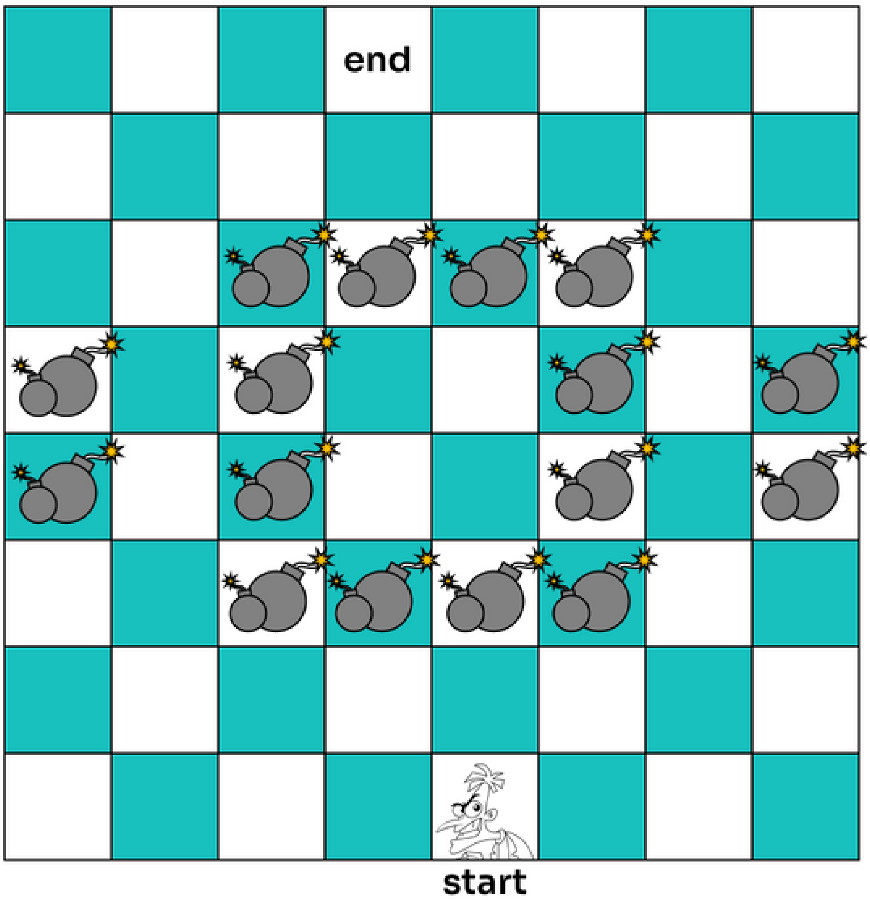
Maximum marks: 15
Minimum marks: -15

Phineas and Ferb unveil the Coolest Coaster Ever to the neighborhood kids. Now all they have to do is to ride the rollercoaster. Within no time, all of them board the rollercoaster and embark on an amazing adventure.
Elsewhere, Agent P tries to free himself from the shackles of Dr.Doofenshmirtz, he grabs Doofenshmirtz's keys (attached to his pocket) with a micro-magnet that he carries everywhere, gets out of the cage, and kicks Doofenshmirtz in his leg.

1. [SINGLE CORRECT ANSWER]
Hurt, Dr.Doofenshmirtz hopped around taking L-shaped jumps (like the knight in chess) around the 8 x 8 floor. How many minimum numbers of jumps should he take to reach the control button from the position he got hurt?

Well obviously he should not jump on the (danger symbol) marked spaces

a) 12
b) 10
c) 8
d) 6



2. [INTEGER TYPE ANSWER]

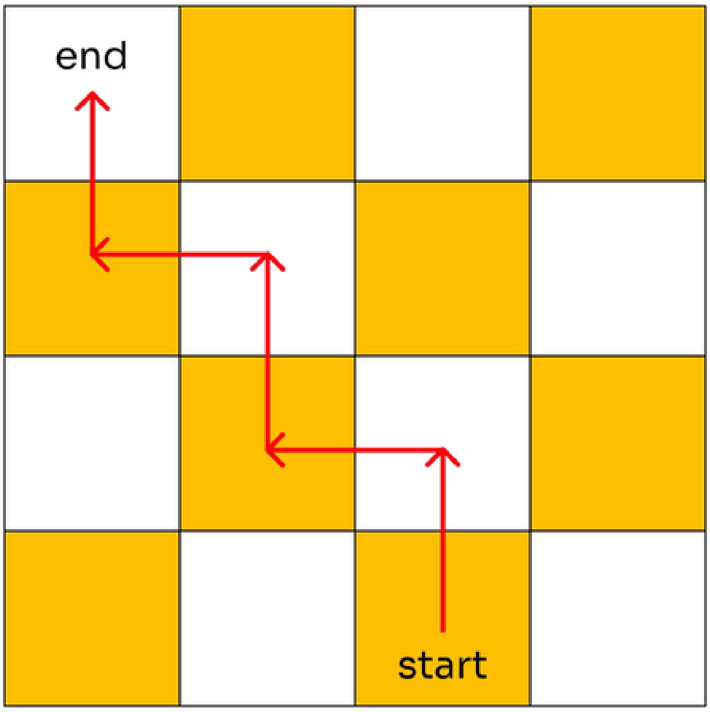
Now, Perry the platypus (Agent P) tries to figure out a way to reach the Magnetism Magnifier (The evil machine on which Dr.Doofenshmirtz is working).

Perry checked it from the corner of his eyes and there was "Magni-Magnifinator" inscribed on the panel……. He finds a huge note on the side of the panel he checks it and weirdly enough both the letters Y and Z are missing in that note.

On the call, Major Monogram tells Perry that this was actually a puzzle that Doofensmirtz gave Vanessa (his daughter) when she was young.

He also found a note in a diary which will help Perry to solve this code

And the note is as follows:

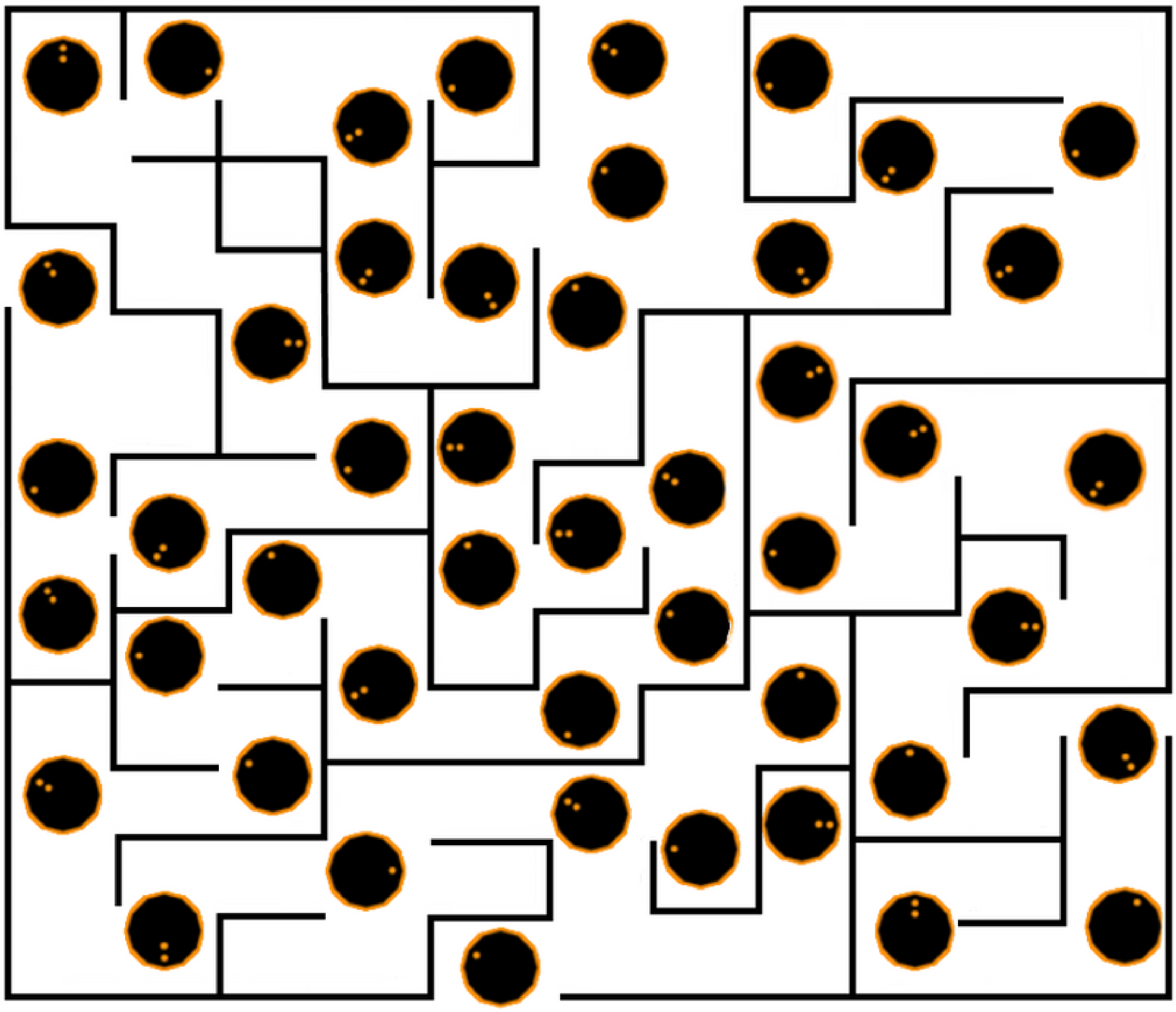


The board is rotated here …
and The pattern follows everywhere

Only one way to connect the start and end…
by which the code is sent

The path is unique… and that's what you seek… you will have to be quick.

But according to the other secret service agent, only one of these paths is correct.



Now, Your task is to help Agent P find the code and report the sum of position 7th and 10th letter of the code according to alphabetical order.
For example in Alphabetical order A is 1, B is 2, C is 3, ... etc.

3. [SINGLE CORRECT ANSWER]
Agent P then makes a permutation $a_1, a_2, a_3, \ldots, a_n$ of the numbers 1,2,3, ... , n so that he couldn't get caught. His permutation is almost sorted if for all i from 1 to n-1, $a_{i+1} \geq a_i - 1$
He, therefore finds the kth lexicographically smallest permutation.
Given n = 15 and k = 2021, If that permutation is $b_1, b_2, \ldots, b_{15}$. Then what is b_{10} ?
*note: Permutation p is lexicographically smaller than a permutation q if and only if the following holds: in the first position where p and q differ, the permutation p has a smaller element than the corresponding element in q.

a) 10 b) 4
c) 3 d) 8

Candace manages to persuade Linda to come along and see the work of Phineas and Ferb. They leave for their home in Linda's car.
On the other side, Agent P defeats Dr.Doofenshmirtz and steals his Magnetic magnifier, attaching it to his Nuclear powered Perry Rocket, and takes it away from Doofenshmirtz.
Here inside the city, The kids reach the broken end of the roller coaster tracks and the coaster derails from the track and moves into the streets of Danville.

4. [SINGLE CORRECT ANSWER]

Candace, the roller coaster, Agent P, and Magnetic magnifier ray move along the edges of a wired 3-dimensional cubic atmosphere of Danville. They can always see each other. If the speed of the Roller coaster, Agent P and the magnetic magnifier ray move is 3 km/hr (Yes it's a Nuclear Powered Rocket😟). By what minimum speed should Agent P move so that he does not get caught in any condition ever in any case?

a) 1
b) 9
c) 5
d) 4

The roller coaster bursts off, sending the kids flying into the air, and they land into the backyard tree of Phineas's home. Phineas and Ferb fall down the tree into their backyard while the other kids are still stuck up there. Here Candace reaches home, and to her surprise, finds them sitting quietly in their backyard. Her mom, on seeing this, greets the boys and scolds Candace for cooking up another story against the 'innocent & quiet' kids. She tells Candace to help her with the groceries and walks away.

The other kids now fall out of the tree, all saying how much they liked the ride. Isabella leaves, saying it really was the best coaster ever. Everyone is finally happy and relieved!

The next day Phineas was singing in his garden and, "I am lost in the rhythmm……". Ferb has an and idea and discusses it with Phineas. Phineas and Ferb then bring their drums and guitar from their warehouse to the backyard. Candace observes this from the corner of her and yells…. "Mommm…Phineas and Ferb are……"
Outro music plays
They are up to something… and more adventures are yet to come!